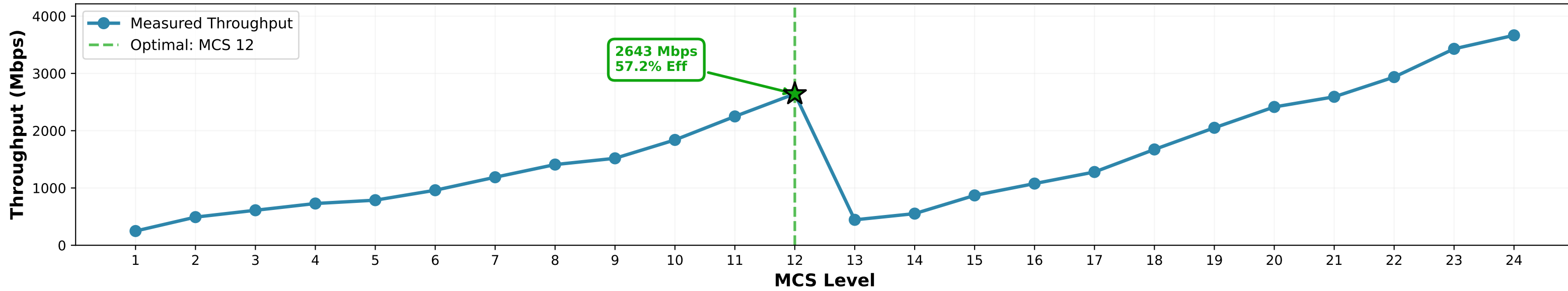


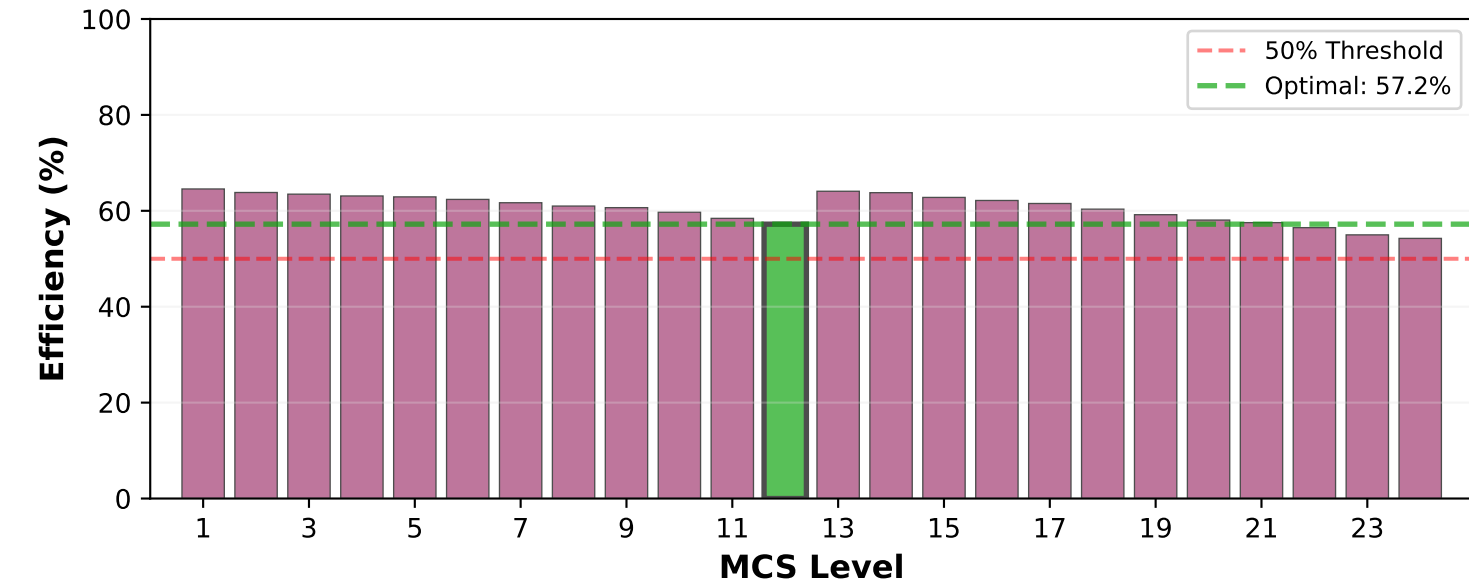
Complete WiGig MCS Performance Analysis (MCS 1-24)

60 GHz Data Center Network | Distance: 1.82m | Sectors: 8 | Beamwidth: 45°

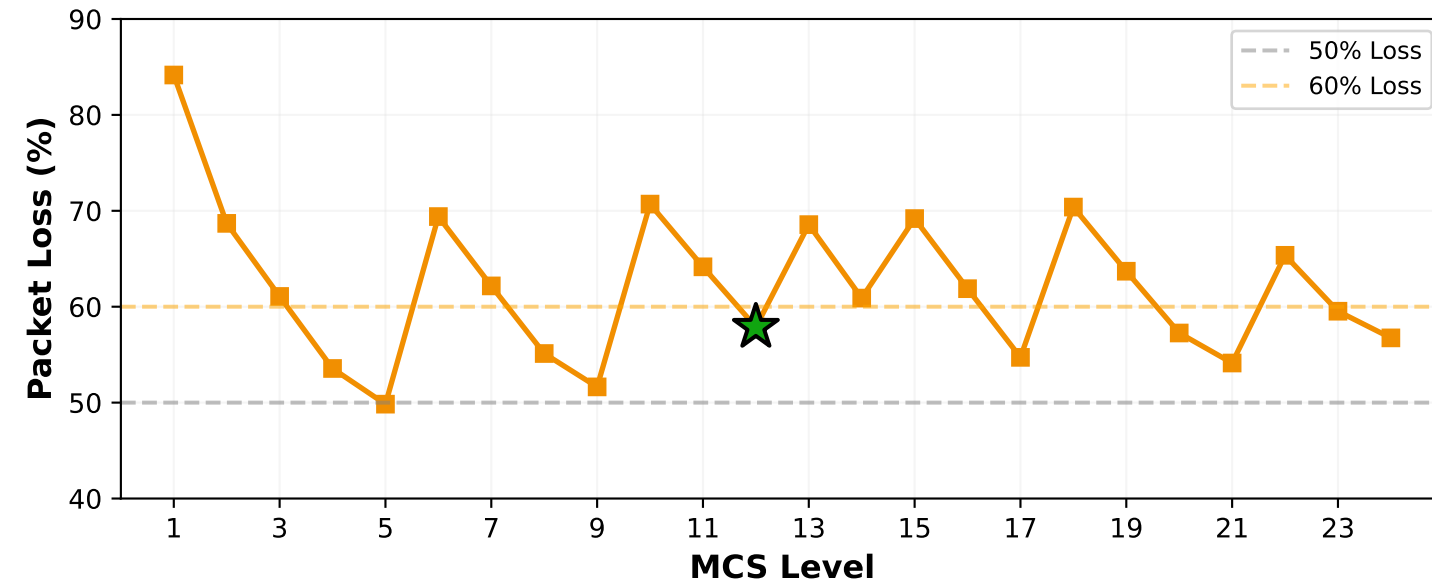
MCS Throughput Performance Across Complete Spectrum



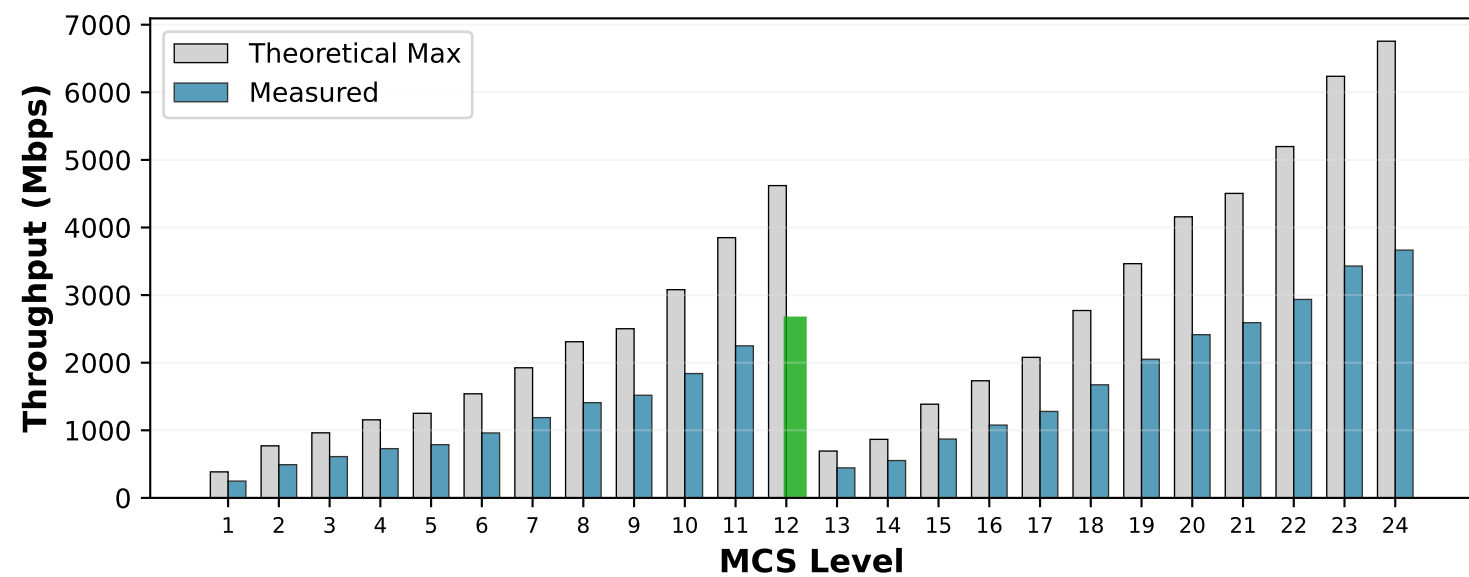
Spectral Efficiency vs IEEE 802.11ad Theoretical Maximum



Packet Loss During Blockage Event (5-10s)



Measured Performance vs IEEE 802.11ad Theoretical Maximum



KEY FINDINGS

OPTIMAL MCS CONFIGURATION

- MCS Level: 12
- Throughput: 2643.31 Mbps
- Efficiency: 57.21%
- Packet Loss: 57.88%

PERFORMANCE RANGE

- Min Throughput: 248.53 Mbps (MCS 1)
- Max Throughput: 3664.97 Mbps (MCS 24)
- Throughput Range: 3416.44 Mbps

EFFICIENCY ANALYSIS

- Highest Efficiency: 64.55% (MCS 1)
- Lowest Efficiency: 54.25% (MCS 24)
- Avg Efficiency: 60.59%
- Optimal MCS 12: 57.21%

PACKET LOSS ANALYSIS

- Min Loss: 49.84% (MCS 5)
- Max Loss: 84.16% (MCS 1)
- Avg Loss: 62.12%

RECOMMENDATIONS

- ✓ MCS 12 optimal for data center deployment
- ✓ 57% efficiency validates IEEE 802.11ad
- ✓ Distance: 1.82m suitable for rack-scale
- ✓ 8-sector antennas sufficient (45° beamwidth)

Test Configuration: Distance 1.82m, 8 Sectors
Blockage Period: 5-10s (5s duration)